

Flight Deck Sentry — Tester Guide

Companion app for the DJI RC Plus · version 0.4.1 · runs in the background while you fly · alerts you to manned aircraft before they get close

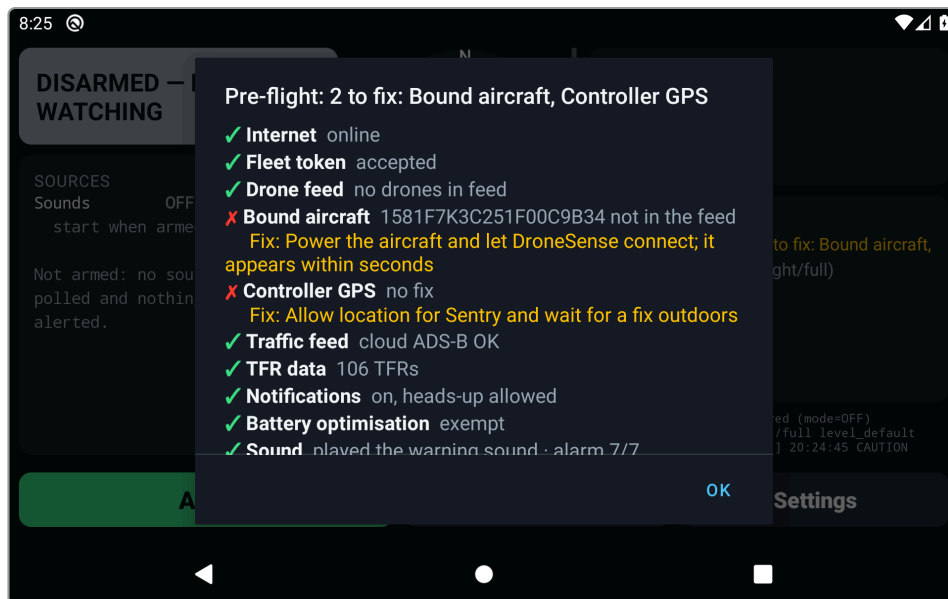
1. What it does (and doesn't)

Sentry watches the airspace around **your drone** and **notifies you** on the controller: a sound, a vibration, and a banner over the flight app. It does not show a map and does not deconflict for you. When it fires, you look at your map or AirSense and decide. It never takes the screen while you're flying, never installs updates while armed, and only ducks the flight app's audio for the second a tone plays.

It knows where your drone is from your organization's drone feed, and where manned aircraft are from public ADS-B (plus a local receiver if your truck has one). If the drone isn't in the feed yet, it protects the space around the controller instead.

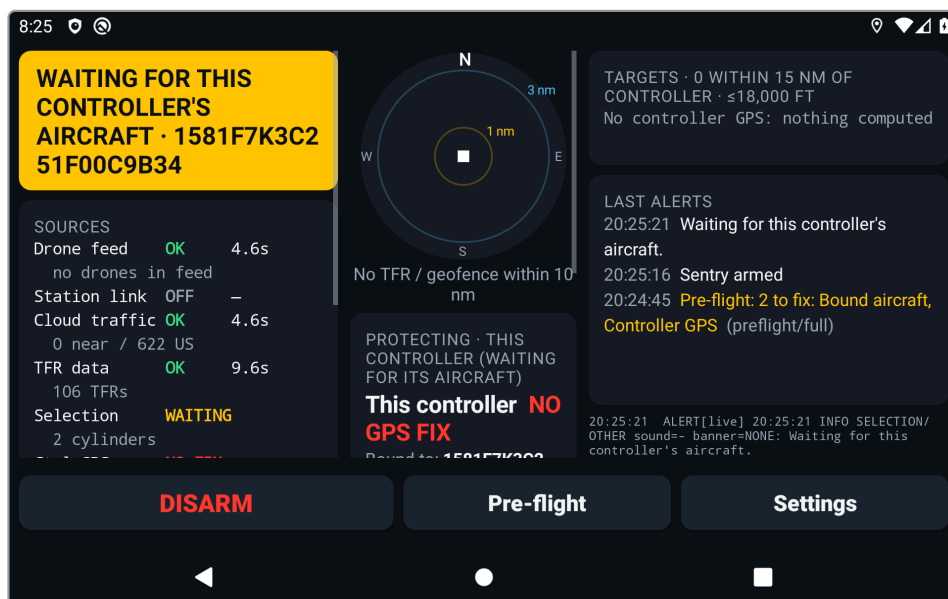
2. Install (5 minutes)

1. On the RC Plus, open the browser and go to github.com/drobinson911/flight-deck-sentry/releases/latest, tap **flight-deck-sentry.apk**. Allow "install unknown apps" when Android asks. (If you had a version older than 0.3.0, uninstall it first.)
2. Open **Flight Deck Sentry** → **Settings**. Paste the **fleet token** (sent to you separately; never share it).
3. **This controller's aircraft:** power your drone with the flight app connected, then tap it in the "aircraft in the feed now" list, or type its serial from the registration card. This is permanent for this controller: Sentry watches this aircraft and no other.
4. Back on the main screen tap **ARM**. Allow notifications and location, and accept the battery-optimisation exemption when asked.
5. Tap **Pre-flight**. Fix anything with a red **X** (each line tells you how). Green all the way down = ready.
6. Switch to your flight app. Sentry keeps running; its banners appear over the flight app.



Pre-flight check: every line is a pass/fail with the fix. Here the aircraft is not powered yet and the controller has no GPS fix.

3. The main screen



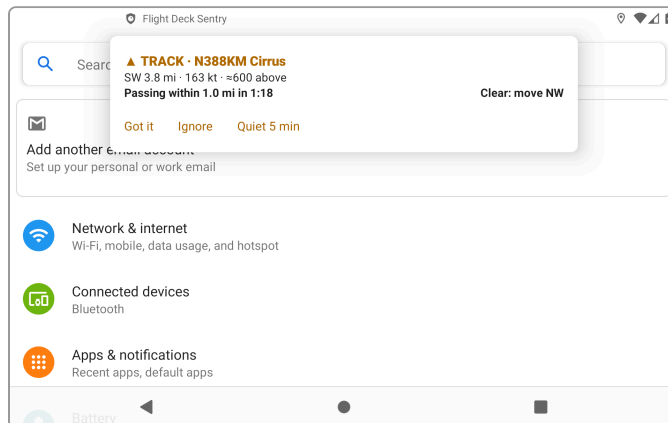
Armed and waiting for the bound aircraft to appear in the feed. Left: sources and their ages. Centre: rings around the drone. Right: targets and the last alerts. The bottom bar never moves.

- **ARM / DISARM:** Sentry only polls and alerts while armed. Disarm after landing.
- **Sources:** each feed shows OK/WAIT and how old its data is. If "Drone feed" says NO TOKEN or TOKEN REJECTED, fix the token in Settings.
- **Bound to:** the serial and callsign Sentry is watching. "Waiting for this controller's aircraft" means it's not in the feed yet.

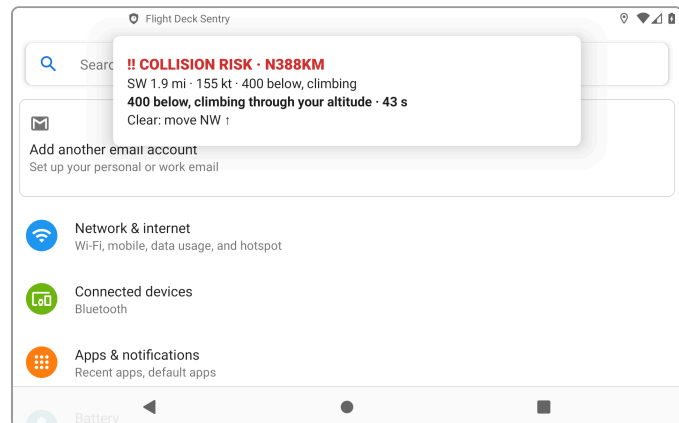
4. What the alerts mean

Alert	When	What you do
▲ TRACK (amber)	An aircraft's current track carries it through your area within 2 minutes. Popup + one sound.	Open your map, find it, start planning. Banner counts down.
⚠ WARNING (red)	Predicted to pass within ½ mile inside 60 s, or already inside the ½-mile ring. Sound + vibrate; repeats faster as it closes (every 20 → 12 → 6 s), softer on repeats.	Act now: move perpendicular to its track (the banner's "Clear:" hint), or climb/descend as advised.
!! COLLISION RISK	Predicted miss under 500 ft sideways and 300 ft vertically within 60 s, or it will cross your altitude while within ½ mile. Continuous tone every 3 s. Cannot be dismissed.	Immediate evasive action.
● PASSING / ○ CLEAR	It's moving away / it's gone.	Carry on.
Zone entry	An aircraft enters the TFR you're in, or a fence you set.	Note it; the cadence above takes over.
Internet lost / regained · aircraft bound / lost	Housekeeping, one sound each.	Check the Sources panel.

Sentry only makes a sound for aircraft **inside its rings AND inside the altitude band** (surface up to 1,500 ft above your drone). Everything else is silent and just shows in the target list. Advisory-level (3 nm) is banner-only by default.



TRACK banner over another app, with the three actions.

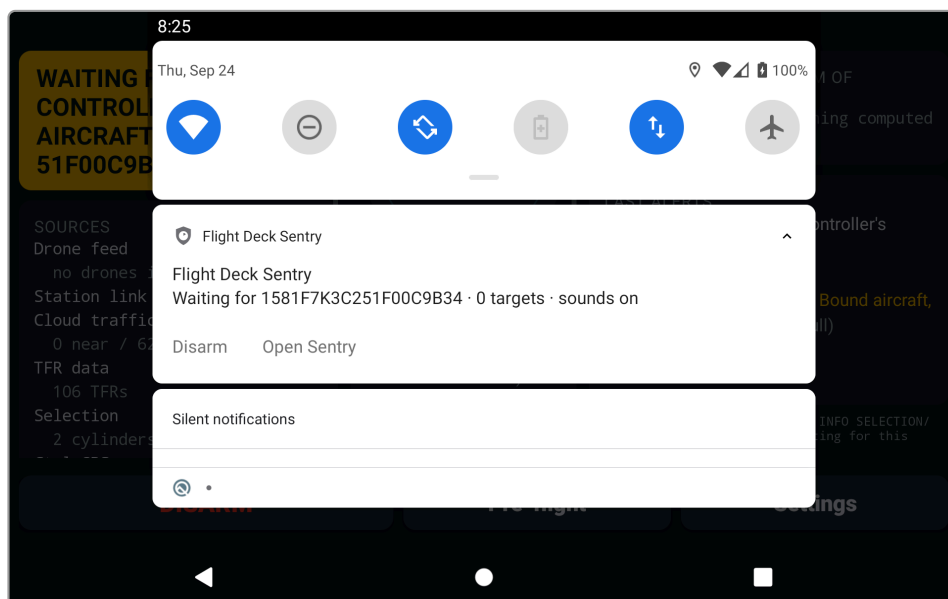


COLLISION RISK: all four lines, no dismiss buttons.

5. Banner buttons (you never need to open the app mid-flight)

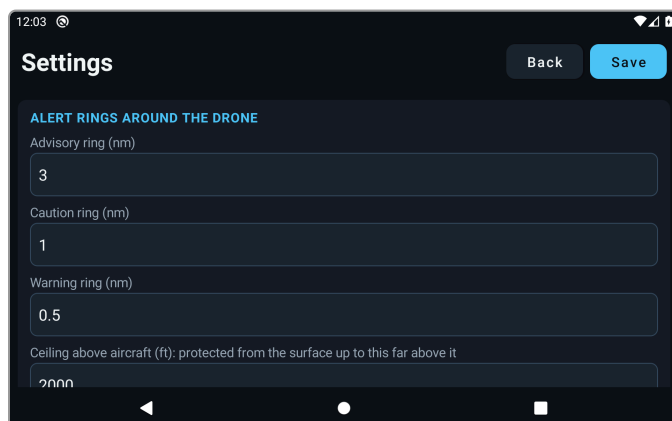
- **Got it:** quiets that aircraft's repeats for 60 s. It comes straight back if the aircraft escalates, turns toward you, or becomes a collision risk.
- **Ignore:** quiets that aircraft until it clears your rings (same escape hatch). For the air attack ship you already know about.
- **Quiet 5 min:** all traffic sounds off for 5 minutes; banners keep updating silently; Sentry says "sounds on" when it ends.

- The small persistent notification has **Disarm** and **Open Sentry**; that's the only way into the app.

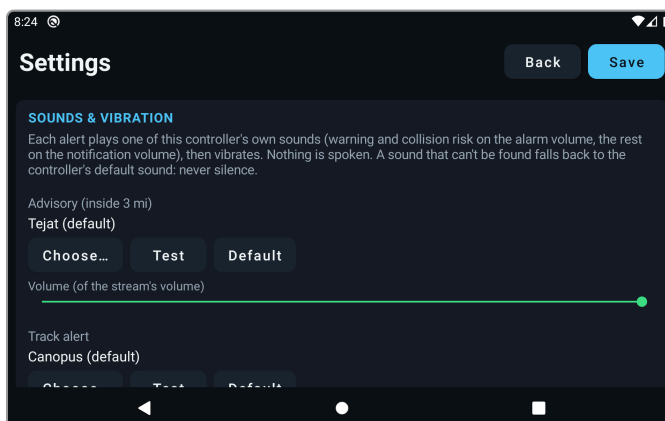


The status notification with Disarm / Open Sentry.

6. Settings you may want to touch



Rings and the altitude band (top of Settings).



Sounds: pick any sound on the controller per alert level, Test each.

- Rings:** 3 / 1 / 0.5 nm by default. **Ceiling above aircraft:** 1,500 ft.
- Cylinders:** what Sentry protects around the controller when your aircraft isn't in the feed.
- Prediction:** track alert 120 s, warning 60 s, collision 60 s. Leave these unless asked.
- Alert style:** Standard / Quiet (Quiet doubles the intervals). **Banner duration:** 5 s.
- Sounds & vibration:** choose per level; Test plays it. If a sound is silent, pick another.
- Every field auto-saves as you type; Enter or tapping outside closes the keyboard.

7. Try it before you fly: the demo replay

Settings → **Replay: demo encounter**. It replays a real recorded encounter (drone "DEMO-1" vs a Cirrus) through the live alert engine: you'll hear TRACK at -86 s, WARNING at -43 s, the closing repeats, PASSING and CLEAR. Tick "synthetic crossing" to hear COLLISION RISK. Nothing about the replay is live traffic.

8. Updating

Settings → **App update** shows installed vs latest and the notes; **Download and install** does the rest (one tap in Android's installer). Sentry checks once a day on its own and posts a quiet notification when there's a new one. It will never install while you're armed.

9. What we need from you (testing)

1. First flight: arm Sentry, fly a normal hop with your flight app in front. Tell us: did any banner cover something important? Did the flight app stutter, drop video, or lose audio? Any sign-outs?
2. Sounds: could you hear the tone over rotor and road noise? Did the vibration register? Which sound did you pick?
3. Alerts: for every TRACK/WARNING you get, note the time and whether it matched what you saw on your map. False alarms and misses are the most useful thing you can send.
4. Battery: note the controller's battery drop over a 30-minute flight with Sentry armed vs not.
5. Send screenshots of anything confusing. Settings → the log at the bottom of the main screen shows every alert with its time.

Sentry is an aid, not a guarantee: it only sees aircraft that broadcast ADS-B (or that your local receiver hears), and it assumes they keep flying straight. See-and-avoid and your VO still come first.